Midterm Exam

ECO372H1F: Data Analysis & Applied Econometrics in Practice

Total: 100 points

Instructions. Answer all questions. Point allocations are indicated in brackets next to each question. Write your answers in the space provided; if you need additional space, continue on the extra sheets at the end of the exam and indicate clearly where the answer is continued. Show all relevant work.

Exam structure:

- **Section A:** Difference-in-Differences (40 points)
- **Section B:** Regression Discontinuity Design (40 points)
- **Section C:** Panel Data Methods (20 points)

Section A: Evaluation of Marijuana Legalization on County Homicide Rates (40 points)

In 2014, several U.S. states legalized recreational marijuana, including California, New York, Oregon, Washington, and Arizona. Other states (e.g., Texas, Georgia, Mississippi, Alabama) maintained recreational prohibition through 2019.

A criminal justice research unit (CJRU) at a federal agency is tasked with evaluating the effect of the 2014 legalization on county-level homicide rates.

Data. The CJRU has access to county-level administrative data containing:

(1) Annual homicide rates per 100,000 residents, 2009–2019;
(2) Each county's state of location and the state's marijuana legalization status;
(3) County-level covariates X_{ct} (population, age structure, median income, unemployment rate, poverty rate).

Setup. For simplicity, restrict the sample to counties in states that legalized marijuana in 2014 (treated, $D_c = 1$) and states that had not legalized by 2019 (comparison, $D_c = 0$). Let Y_{ct} denote the homicide rate in county c in year t. Potential outcomes are $Y_{ct}(1)$ and $Y_{ct}(0)$, with the observed outcome following the standard switching equation.

Question A.1. [10]

The CJRU first considers estimating the causal effect of marijuana legalization using a cross-sectional OLS regression on post-legalization (2014) data:

$$Y_{c,2014} = \alpha + \theta D_c + X_{c,2014}\beta + u_c$$

Write down the formula for selection bias in this setting, and explain how selection bias might arise in the research design laid out. Provide a plausible example of an omitted factor that would generate this bias and discuss its likely sign.

Question A.2. [5]

Concerned about the limitations of OLS, the CJRU decides instead to use a difference-in-differences (DiD) design comparing homicide changes between 2013 (pre-legalization) and 2014 (post-legalization) in legalizing vs. non-legalizing counties.

Explain why this strategy is (in principle) a Difference-in-Differences design.

(NB: No need to write down the empirical model here; see next question.)

Question A.3. [10]

Write down the DiD empirical model the CJRU should estimate to recover the ATT of legalization on homicides in 2014. Discuss the conditions under which it yields unbiased (consistent) estimates of the treatment effect δ, and explain how this strategy addresses the selection-bias problem you identified in Question A.1. Based on Figure 1, are the conditions satisfied?

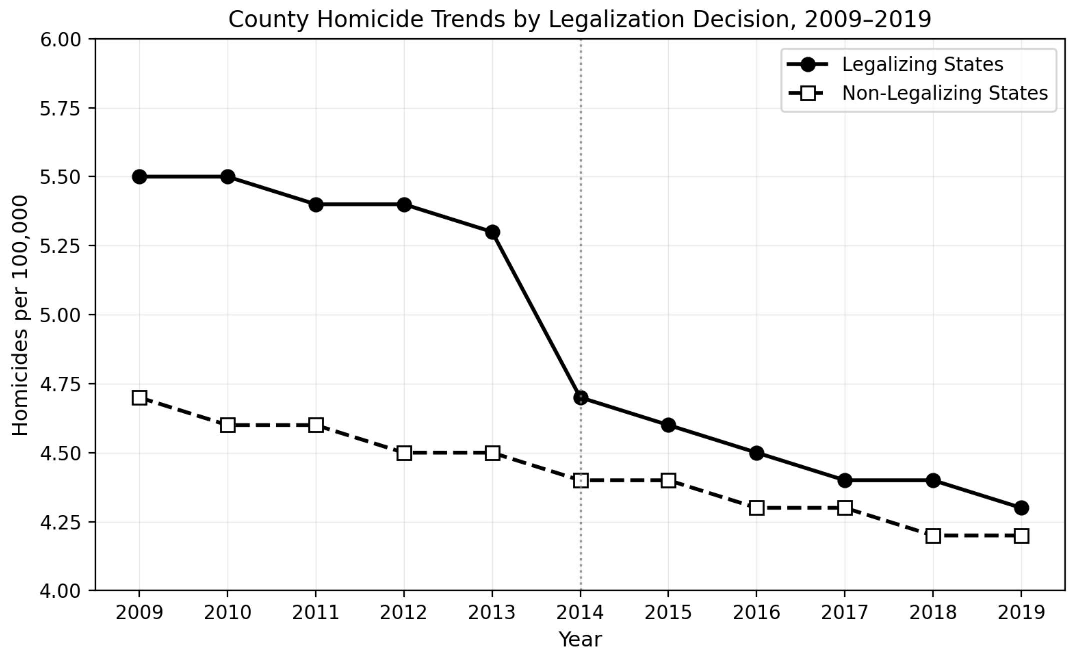

Figure 1: County Homicide Trends by Legalization Decision, 2009–2019.

Question A.4. [15]

The CJRU generates the following population-weighted estimates from the 2×2 DiD comparison (see Table 1) and a plot of county homicide trends by legalization decision (see Figure 1).

With reference to your answers to the previous questions in this section, critically discuss the results. In your role as analyst, emphasize the **interpretation of the results**, **threats to identification**, and **necessary tests** to verify the validity of the design for purposes of this evaluation.

Table 1: Population-Weighted Average County Homicide Rates (per 100,000 residents).

	Legalizing	Non-Legalizing	Gap
2013	5.3	4.5	0.8
2014	4.7	4.4	0.3
Trend (2014 – 2013)	−0.6	−0.1	**−0.5**
			(0.18)
			[$p = 0.005$]

Notes: Population-weighted county-level homicide rates among adult residents. Sample includes 815 counties in states that legalized marijuana in 2014 and 1,124 counties in states that had not legalized by 2019. Standard error of the DiD estimate in parentheses; p-value in brackets. Standard errors clustered at the state level.

Section B: Collective Property Rights and Deforestation in the Brazilian Amazon (40 points)

The Amazon rainforest holds roughly half of the world's remaining tropical forest, and its preservation is central to global climate-change mitigation. Sixty percent of the Amazon lies within Brazil's "Legal Amazon," where rising deforestation—driven largely by the clearing and burning of land for agriculture—has become a serious policy concern.

A substantial share of the forest that remains intact sits within **indigenous territories**. In Brazil, an indigenous territory acquires *full property rights* only at the end of a multi-step legal process called **demarcation**. The final step, **homologation**, is a presidential decree that registers the territory in the national land registry. Only after homologation does the territory become the permanent possession of its indigenous community: no third party may contest it, the federal government becomes constitutionally responsible for protecting it, and monitoring and enforcement agencies (which respond to satellite-detected forest clearing in near-real time) begin operating on its behalf. Before homologation, a territory may have its boundaries drawn and mapped, but it holds none of these legal protections.

You have been asked to evaluate the following question: **does granting full property rights (homologation) reduce deforestation inside indigenous territories?** You have access to satellite data measuring the percent of forest cover lost ("percent deforestation") in each small plot ("pixel," roughly $4\,\text{km} \times 4\,\text{km}$) of land, every year from 1982 to 2016. For each pixel you also observe whether it lies inside or outside an indigenous territory, the territory's legal status, and the year (if any) in which the territory was homologated.

Prior research has established that deforestation is strongly predicted by a plot's *geography and accessibility*—in particular, its proximity to roads, rivers, and mines, its elevation, and its rainfall—since these determine how cheaply forested land can be reached and converted to agricultural use. Indigenous territories are not located at random with respect to these characteristics.

It is useful to distinguish three groups of land in the data:

- **Non-homologated territories** — land identified and mapped as indigenous, but never granted full property rights;
- **Before homologation** — land in territories that *would* eventually be homologated, observed in the years *before* rights were granted;
- **After homologation** — the same territories, observed in the years *after* rights were granted.

B.1 — The naive comparison [10]

A first analyst proposes estimating the effect of property rights with the following cross-sectional regression, pooling all pixels:

$$\text{Defor}_i \;=\; \alpha \;+\; \beta\,\text{FullRights}_i \;+\; \varepsilon_i,$$

where Defor_i is percent deforestation in pixel i and $\text{FullRights}_i = 1$ if pixel i lies inside a homologated (full-property-rights) territory, and 0 otherwise.

Question B.1.1. Identify **at least one** omitted variable that would bias the OLS estimate $\hat{\beta}_{OLS}$. State clearly: (i) why the variable belongs in the error term ε_i, (ii) how it is correlated with FullRights_i, and (iii) the resulting *direction* of the bias in $\hat{\beta}_{OLS}$—that is, whether the naive estimate would overstate or understate the true protective effect of property rights. [10]

B.2 — A regression discontinuity design [20]

To address the problem in B.1, the analyst now restricts attention to pixels lying within a narrow band ("buffer") on either side of an indigenous territory's border, and defines the **running variable** X_i as the distance from pixel i to the nearest border (negative outside the territory, positive inside, zero at the border).

Question B.2.1. Explain *in words* why comparing pixels just inside the border to pixels just outside is, in principle, a Regression Discontinuity Design. How does this comparison address the selection problem you identified in B.1? (*No equations required.*) [5]

Question B.2.2. Is this a **sharp** or a **fuzzy** RDD? Justify your answer with reference to how treatment (full property rights) is assigned at the border. [3]

Question B.2.3. The RDD estimator focuses on pixels right at the border between homologated territories and the surrounding land. Are these results applicable to deforestation **deep in the interior** of a large homologated territory? Why or why not? [4]

Question B.2.4. Figure 2 below plots percent deforestation against distance to the border for each of the three groups defined earlier in this section, with a fitted curve on each side of the cutoff.

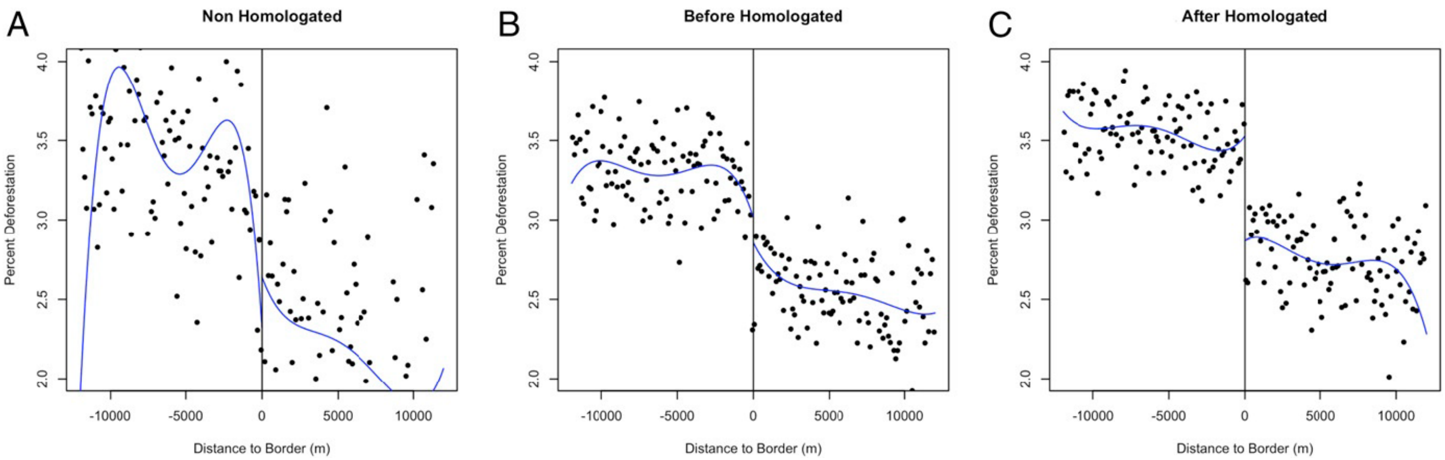

Figure 2: Regression discontinuity plots for (A) non-homologated territories, (B) homologated territories before homologation, and (C) homologated territories after homologation. Dependent variable is percent deforestation. Running variable is distance to the border.

Using Figure 2 to guide your choice, **write down one empirical specification** you would estimate to recover the effect of property rights. Your answer must make clear: [8]

(i) *which subsample* (which panel of Figure 2) you would estimate on, and why;

(ii) *what functional form* you would fit on each side of the cutoff;

(iii) *which coefficient* in your specification captures the effect of property rights on deforestation, and how that coefficient should be **interpreted** (one or two sentences); and

(iv) the **identifying assumption** under which this coefficient consistently estimates the effect of property rights.

B.3 — Interpreting the results and assessing threats to identification [10]

Table 2 below reports the RDD estimates from the paper for the **after-homologation** subsample, across four functional forms.

Table 2: Regression discontinuity estimates: Effect of indigenous property rights on deforestation, after-homologation subsample.

	Linear	Quadratic	Cubic	4th order
RDD estimate	-2.07^{**}	-2.66^{**}	-2.48^{**}	-1.99^{**}
	(0.024)	(0.033)	(0.039)	(0.035)

Notes: Dependent variable is percent of deforestation. Running variable is distance to the indigenous territory border in meters. Robust standard errors clustered at the indigenous territory level are in parentheses. $^{**}P < 0.01$, $^{*}P < 0.05$.

Question B.3.1. Locate the column in Table 2 that most closely corresponds to the specification you wrote down in Question B.2.4. Interpret this estimate. Your answer should discuss the **magnitude** (including what it implies substantively, given that baseline deforestation is around 3%) and the **statistical significance** of the result. [5]

Question B.3.2. Identify **one potential threat** to the RD design used in this study. (It need *not* be a valid threat for this specific application—any plausible RDD threat from class is acceptable.) Then discuss: [5]

(i) *intuitively*, whether this threat is likely to matter in this particular setting, given that the running variable is the geographic distance to an indigenous territory's border; and

(ii) what **test or diagnostic** you could run on the data to assess whether the threat is actually present.

Section C: Panel Data Methods (20 points)

Question C.1 — Why pooled OLS is biased [10]

Consider the following panel regression of firm-level productivity on R&D spending across firms i and years t:

$$\text{productivity}_{it} = \beta_0 + \beta_1 \text{R\&D}_{it} + \alpha_i + e_{it}$$

where α_i is a firm-specific unobserved effect (e.g., managerial talent or organizational culture).

(a) If we ignore α_i and run pooled OLS, show the decomposition of the composite error term and explain why $\hat{\beta}_1^{POLS}$ will be biased.

(b) Provide intuitive reasoning for why managerial talent (α_i) and R&D spending would be correlated. What is the likely sign of the bias?

Question C.2 — The random effects assumption [10]

(a) State the random effects assumption and explain in plain English what it requires.

(b) Give one example of a setting where this assumption is plausible, and one where it is implausible. Briefly justify each.

— *End of Exam* —

Extra space